Digital Logic Circuit Design Assignment 1

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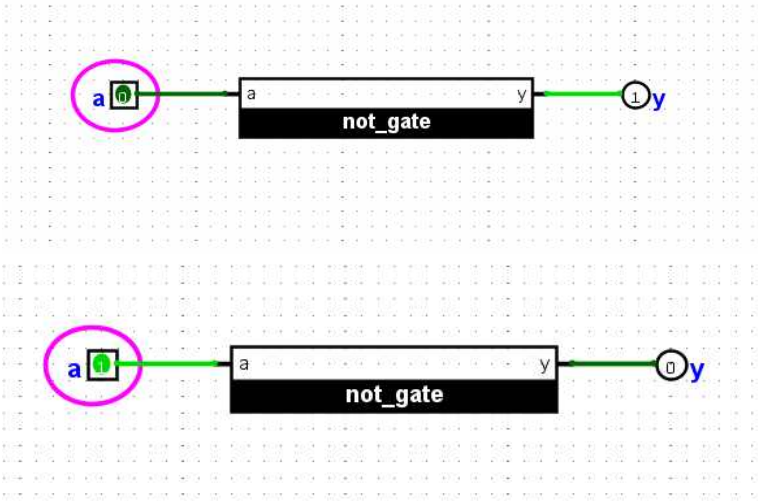
## 1. VHDL Code

```
library ieee;
use ieee.std_logic_1164.all;

entity not_gate is
    port (
        a : in std_logic;
        y : out std_logic
    );
end entity not_gate;

architecture behavioral of not_gate is
begin
    y <= not a;
end architecture behavioral;
```

## 2. Circuit & Simulation Screenshots




## 3. Explanation of Results

The NOT gate was simulated in Logisim-evolution using Questa. When the input a was set to 0, the output y was 1. When the input a was changed to 1, the output y changed to 0. In both cases, the output was the opposite of

the input, matching the truth table of a NOT gate. These results confirm that the VHDL code and the connected circuit correctly perform the NOT operation.